

Skim milk, whey, and casein increase body weight and whey and casein increase the plasma C-peptide concentration in overweight adolescents.

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In adults, dietary protein seems to induce weight loss and dairy proteins may be insulinotropic. However, the effect of milk proteins in adolescents is unclear. The objective was to test whether milk and milk proteins reduce body weight, waist circumference, homeostatic model assessment, plasma insulin, and insulin secretion estimated as the plasma C-peptide concentration in overweight adolescents. Overweight adolescents ($n = 203$) aged 12-15 y with a BMI of 25.4 ± 2.3 kg/m² (mean $\pm$ SD) were randomized to 1 L/d of skim milk, whey, casein, or water for 12 wk. All milk drinks contained 35 g protein/L. Before randomization, a subgroup of adolescents ($n = 32$) was studied for 12 wk before the intervention began as a pretest control group. The effects of the milk-based test drinks were compared with baseline (wk 0), the water group, and the pretest control group. Diet and physical activity were registered. Outcomes were BMI-for-age Z-scores (BAZs), waist circumference, plasma insulin, homeostatic model assessment, and plasma C-peptide. We found no change in BAZ in the pretest control and water groups, whereas it was greater at 12 wk in the skim milk, whey, and casein groups compared with baseline and with the water and pretest control groups. The plasma C-peptide concentration increased from baseline to wk 12 in the whey and casein groups and increments were greater than in the pretest control ($P < 0.02$). There were no significant changes in plasma C-peptide in the skim milk or water group. These data suggest that high intakes of skim milk, whey, and casein increase BAZs in overweight adolescents and that whey and casein increase insulin secretion. Whether the effect on body weight is primary or secondary to the increased insulin secretion remains to be elucidated.